What is Software Engineering?

IEEE defines software as: ”Software is a collection of computer programs, together with data, procedures, rules, and associated documentation, which operate in a specified environment with certain constraints to provide the desired outcomes.”

Software Engineering is the process of designing, developing, testing, and maintaining software. It is a systematic and disciplined approach to software development that aims to create high-quality, reliable, and maintainable software.

1. Software engineering includes a variety of techniques, tools, and methodologies, including requirements analysis, design, testing, and maintenance.
1. It is a rapidly evolving field, and new tools and technologies are constantly being developed to improve the software development process.
1. By following the principles of software engineering and using the appropriate tools and methodologies, software developers can create high-quality, reliable, and maintainable software that meets the needs of its users.
1. Software Engineering is mainly used for large projects based on software systems rather than single programs or applications.
1. The main goal of Software Engineering is to develop software applications for improving quality, budget, and time efficiency.
1. Software Engineering ensures that the software that has to be built should be consistent, correct, also on budget, on time, and within the required requirements.

Key Principles of Software Engineering

1. **Modularity**: Breaking the software into smaller, reusable components that can be developed and tested independently.
1. **Abstraction**: Hiding the implementation details of a component and exposing only the necessary functionality to other parts of the software.
1. **Encapsulation**: Wrapping up the data and functions of an object into a single unit, and protecting the internal state of an object from external modifications.
1. **Reusability**: Creating components that can be used in multiple projects, which can save time and resources.
1. **Maintenance**: Regularly updating and improving the software to fix bugs, add new features, and address security vulnerabilities.
1. **Testing**: Verifying that the software meets its requirements and is free of bugs.

1. **Design Patterns**: Solving recurring problems in software design by providing templates for solving them.
1. **Agile methodologies:** Using iterative and incremental development processes that focus on customer satisfaction, rapid delivery, and flexibility.
1. **Continuous Integration & Deployment:** Continuously integrating the code changes and deploying them into the production environment.

Main Attributes of Software Engineering

Software Engineering is a systematic, disciplined, quantifiable study and approach to the design, development, operation, and maintenance of a software system. There are four main Attributes of Software Engineering.

1. **Efficiency:** It provides a measure of the resource requirement of a software product efficiently.
1. **Reliability:** It assures us that the product will deliver the same results when used in similar working environment.
1. **Reusability:** This attribute makes sure that the module can be used in multiple applications.
1. **Maintainability:** It is the ability of the software to be modified, repaired, or enhanced easily with changing requirements.

Dual Role of Software

There is a dual role of software in the industry. The first one is as a product and the other one is as a vehicle for delivering the product. We will discuss both of them.

1. As a Product

- It delivers computing potential across networks of Hardware.
- It enables the Hardware to deliver the expected functionality.
- It acts as an information transformer because it produces, manages, acquires, modifies, displays, or transmits information.

2. As a Vehicle for Delivering a Product

- It provides system functionality (e.g., payroll system).
- It controls other software (e.g., an operating system).
- It helps build other software (e.g., software tools).

Objectives of Software Engineering

1. **Maintainability:** It should be feasible for the software to evolve to meet changing requirements.
1. **Efficiency:** The software should not make wasteful use of computing devices such as memory, processor cycles, etc.
1. **Correctness:** A software product is correct if the different requirements specified in the SRS Document have been correctly implemented.
1. **Reusability:** A software product has good reusability if the different modules of the product can easily be reused to develop new products.
1. **Testability:** Here software facilitates both the establishment of test criteria and the evaluation of the software concerning those criteria.
1. **Reliability:** It is an attribute of software quality. The extent to which a program can be expected to perform its desired function, over an arbitrary time period.
1. **Portability:** In this case, the software can be transferred from one computer system or environment to another.
1. **Adaptability:** In this case, the software allows differing system constraints and the user needs to be satisfied by making changes to the software.
1. **Interoperability:** Capability of 2 or more functional units to process data cooperatively.

What Careers Are There in Software Engineering?

A degree in software engineering and relevant experience can be utilized to explore several computing job choices. Software engineers have the opportunity to seek well-paying careers and professional progress, although their exact possibilities may vary depending on their particular school, industry, and region.

Following are the job choices in software engineering:

- SWE (Software Engineer)
- SDE (Software Development Engineer)
- Web Developer
- Quality Assurance Engineer
- Web Designer
- Software Test Engineer
- Cloud Engineer ·
- Front-End Developer

- Back-End Developer
- DevOps Engineer.
- Security Engineer

What Tasks do Software Engineers do?

The main responsibility of a software engineer is to develop useful computer programs and applications. Working in teams, you would complete various projects and develop solutions to satisfy certain customer or corporate demands.

Some of the key responsibilities of software engineer are:

- **Requirement Analysis**: Collaborating with stakeholders to understand and gather the requirements to design and develop software solutions.
- **Design and Development**: Creating well-structured, maintainable code that meets the functional requirements and adheres to software design principles.
- **Testing and Debugging**: Writing and conducting unit tests, integration tests, and debugging code to ensure software is reliable and bug-free.
- **Code Review**: Participating in code reviews to improve code quality, ensure adherence to standards, and facilitate knowledge sharing among team members.
- **Maintenance**: Updating and maintaining existing software systems, fixing bugs, and improving performance or adding new features.
- **Documentation**: Writing clear documentation, including code comments, API documentation, and design documents to help other engineers and future developers understand the system.

How Much do Software Engineers Make?

The majority of software developers globally make less than $50,000 annually on average. This covers not only junior developers but also some senior ones based on where they work and for what company. However, it is more likely that a mid-level or senior software engineer makes between $50,000 and $120,000 per year.

Salaries for Careers with a Software Engineering Degree:

Role	**Average Salary in USA**	**Average Salary in Australia**	**Average Salary in India**	**Average Salary in Germany**
Software Engineering	$124,200	€67,260	*₹9,00,000*	€56,787

Backend Dev	$157,602	€72,354	**₹9,61,589**	€60,100
Frontend Dev	$113,249	€69,173	**₹6,50,000**	€49,380
Full Stack Dev	$123,809	€72,286	*₹ 9,20,000*	€51,074
Java Dev	$92,565	€70,958	*₹7,36,651*	€50,635
Python Dev	$123,663	€84,296	*₹5,50,000*	€55,235

What Skills do Software Engineers Need?

To achieve success, software engineers require a unique set of technical and soft skills. These skills include computer programming knowledge for designing, developing, testing, and debugging software, as well as soft skills for interacting with stakeholders, team members, and company leadership.

Following are some must have technical skills to become Software Engineers:

- Coding and computer programming
- Software testing
- Object-oriented design (OOD)
- Software development

Following are some must have soft skills to become Software Engineers:

- Communication skills
- Team player
- Problem solving
- Attention to detail

Program vs Software Product

Parameters	**Program**	**Software Product**

Definition	A program is a set of instructions that are given to a computer in order to achieve a specific task.	Software is when a program is made available for commercial business and is properly documented along with its licensing. Software Product = Program + Documentation + Licensing.
Stages Involved	Program is one of the stages involved in the development of the software.	Software Development usually follows a life cycle, which involves the feasibility study of the project, requirement gathering, development of a prototype, system design, coding, and testing.

Advantages of Software Engineering

There are several advantages to using a systematic and disciplined approach to software development, such as:

1. **Improved Quality:** By following established software engineering principles and techniques, the software can be developed with fewer bugs and higher reliability.
1. **Increased Productivity:** Using modern tools and methodologies can streamline the development process, allowing developers to be more productive and complete projects faster.
1. **Better Maintainability:** Software that is designed and developed using sound software engineering practices is easier to maintain and update over time.
1. **Reduced Costs:** By identifying and addressing potential problems early in the development process, software engineering can help to reduce the cost of fixing bugs and adding new features later on.
1. I**ncreased Customer Satisfaction:** By involving customers in the development process and developing software that meets their needs, software engineering can help to increase customer satisfaction.
1. **Better Team Collaboration:** By using Agile methodologies and continuous integration, software engineering allows for better collaboration among development teams.

1. **Better Scalability**: By designing software with scalability in mind, software engineering can help to ensure that software can handle an increasing number of users and transactions.

1. **Better Security:** By following the Software Development Life Cycle (SDLC) and performing security testing, software engineering can help to prevent security breaches and protect sensitive data.

In summary, software engineering offers a structured and efficient approach to software development, which can lead to higher-quality software that is easier to maintain and adapt to changing requirements. This can help to improve customer satisfaction and reduce costs, while also promoting better collaboration among development teams.

Disadvantages of Software Engineering

While Software Engineering offers many advantages, there are also some potential disadvantages to consider:

1. **High upfront costs:** Implementing a systematic and disciplined approach to software development can be resource-intensive and require a significant investment in tools and training.

1. **Limited flexibility:** Following established software engineering principles and methodologies can be rigid and may limit the ability to quickly adapt to changing requirements.

1. **Bureaucratic**: Software Engineering can create an environment that is bureaucratic, with a lot of processes and paperwork, which may slow down the development process.

1. **Complexity**: With the increase in the number of tools and methodologies, software engineering can be complex and difficult to navigate.

1. **Limited creativity:** The focus on structure and process can stifle creativity and innovation among developers.

1. **High learning curve:** The development process can be complex, and it requires a lot of learning and training, which can be challenging for new developers.

1. **High dependence on tools:** Software engineering heavily depends on the tools, and if the tools are not properly configured or are not compatible with the software, it can cause issues.

1. **High maintenance**: The software engineering process requires regular maintenance to ensure that the software is running efficiently, which can be costly and time-consuming.